

Physics Reference Notes

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1 Similarity transformations of operators: computing $e^A B e^{-A}$

1.1 Motivation

Similarity transformations $B \mapsto e^A B e^{-A}$ are ubiquitous whenever a Hamiltonian is brought to a simpler (often diagonal) form: Bogoliubov transformations of quadratic Hamiltonians, displacement/squeezing of bosonic modes, polaron (Lang-Firsov) transformations, Schrieffer-Wolff transformations, and passage to the interaction picture or a rotating frame are all instances of this construction. If $A^\dagger = -A$ (generator anti-Hermitian), $U = e^A$ is unitary and the transformation preserves Hermiticity and spectra. This is the case in all examples below.

1.2 Method A: The Hadamard lemma

Hadamard Lemma. *For any two operators A, B ,*

$$e^A B e^{-A} = \sum_{n=0}^{\infty} \frac{1}{n!} \text{ad}_A^n(B), \quad \text{with} \quad (1)$$

$$\text{ad}_A^n(B) \equiv [A, B]_n \equiv [A, [A, B]_{n-1}], \quad \text{ad}_A^0(B) \equiv [A, B]. \quad (2)$$

This series is especially helpful whenever the nested commutator yields zero at for a finite n or if the nested commutators eventually reproduce a small, recognizable pattern.

Strategy 1: Reduce to elementary operators. Utilize that for any A, X, Y ,

$$e^A (XY) e^{-A} = (e^A X e^{-A}) (e^A Y e^{-A}). \quad (3)$$

It thus suffices to compute $e^A a e^{-A}$ and $e^A a^\dagger e^{-A}$ once and any polynomial in these follows by substitution and normal-ordering, with no further Hadamard series needed.

Strategy 2: The dagger shortcut. If $A^\dagger = -A$, $U = e^A$ is unitary with $U^\dagger = e^{-A} = U^{-1}$, so $e^A B e^{-A} = U B U^\dagger$. Therefore, taking the dagger simply yields

$$(e^A B e^{-A})^\dagger = e^A B^\dagger e^{-A}. \quad (4)$$

Once $e^A a e^{-A}$ is known, $e^A a^\dagger e^{-A}$ follows for free by Hermitian conjugation, roughly halving the bookkeeping.

1.3 Method B: The generator flow method (not proof read since not used yet)

Define $X(\theta) \equiv e^{\theta A} X e^{-\theta A}$, $X(0) = X$. Differentiating,

$$\frac{dX(\theta)}{d\theta} = [A, X(\theta)] = \text{ad}_A(X(\theta)). \quad (5)$$

This ordinary differential equation (ODE) is exactly equivalent to Method A – the Hadamard series is its formal Taylor solution around $\theta = 0$ – but is can be more tractable in practice.

Key fact: closure $\Rightarrow$ finite-dimensional linear ODE. If ad_A maps a finite set of operators $\{X_1, \dots, X_n\}$ into linear combinations of themselves, the operator ODE collapses to a matrix ODE:

$$\frac{d}{d\theta} \vec{X}(\theta) = M \vec{X}(\theta), \quad [A, X_j] = \sum_i M_{ij} X_i, \quad \vec{X}(\theta) = e^{\theta M} \vec{X}(0), \quad (6)$$

reducing the problem to exponentiating an ordinary $n \times n$ matrix.

1.4 When to prefer which method

Both methods are mathematically identical. Method A is fast when the pattern is recognizable by eye; Method B is more systematic for anything beyond the simplest single-operator case, especially with mixed or multiple generators to track jointly.

1.5 Scope: when this machinery fails

The closure property requires A to be at most quadratic in a set of canonically (anti-) commuting operators. For a genuinely quartic operator such as $H_{\text{int}} = U \sum_i (n_i - \frac{1}{2})(n_{i+1} - \frac{1}{2})$, $\text{ad}_A(H_{\text{int}})$ does not close on a finite set for generic one-body A (e.g. $A \propto J$ or $A \propto T$) – $[J, H_{\text{int}}]$ is generically nonzero and does not reduce to a small closed set of operators. The Hadamard series then cannot be resummed in closed form; one is limited to a perturbative treatment in the strength of A .

1.6 References

- J.J. Sakurai & J. Napolitano, *Modern Quantum Mechanics*.
- C. Gerry & P. Knight, *Introductory Quantum Optics*.
- D.F. Walls & G.J. Milburn, *Quantum Optics*.
- J.-P. Blaizot & G. Ripka, *Quantum Theory of Finite Systems*, MIT Press.
- A. Perelomov, *Generalized Coherent States and Their Applications*, Springer.
- J.W. Negele & H. Orland, *Quantum Many-Particle Systems*.
- B.C. Hall, *Lie Groups, Lie Algebras, and Representations*.

2 The tensor product and the Kronecker product as one of its possible representations

2.1 The tensor product of vector spaces

Let $\underline{v} = \{v_i \mid i \in I\}$ and $\underline{w} = \{w_j \mid j \in J\}$ be bases for V and W . Then $V \otimes W$ is the product space, where there exists a basis which can be identified uniquely with the pairs of the cartesian product

$$\underline{v} \times \underline{w} = \{(v_i, w_j) \mid i \in I, j \in J\}, \quad (7)$$

so that

$$\dim(V \otimes W) = \dim V \cdot \dim W. \quad (8)$$

Since $V \otimes W$ is a vector space, an arbitrary element of it can be written as

$$\sum_{(i,j) \in I \times J} c_{ij} (v_i \otimes w_j). \quad (9)$$

The bilinear map $\otimes$ is defined by

$$\otimes: V \times W \longrightarrow V \otimes W, \quad (10)$$

$$(v, w) \longmapsto v \otimes w = \sum_{(i,j) \in I \times J} a_i b_j (v_i \otimes w_j), \quad (11)$$

for $v = \sum_i a_i v_i$ and $w = \sum_j b_j w_j$. As a check of bilinearity, writing $v' = \sum_i a'_i v_i$ and $v'' = \sum_i a''_i v_i$,

$$\begin{aligned} (v' + v'') \otimes w &= \left(\sum_i a'_i v_i + \sum_i a''_i v_i \right) \otimes w = \left(\sum_i (a'_i + a''_i) v_i \right) \otimes w \\ &= \sum_{i,j} (a'_i + a''_i) b_j (v_i \otimes w_j) \\ &= \sum_{i,j} a'_i b_j (v_i \otimes w_j) + \sum_{i,j} a''_i b_j (v_i \otimes w_j) \\ &= v' \otimes w + v'' \otimes w, \end{aligned} \quad (12)$$

and analogously in the second argument.

2.2 The tensor product of operators

Introduce operators $A: V \rightarrow V$ and $B: W \rightarrow W$ with V, W vector spaces. How is $A \otimes B: V \otimes W \rightarrow V \otimes W$ given? Define a map $\beta: V \times W \rightarrow V \otimes W$, $\beta(v, w) := Av \otimes Bw$. It is bilinear, e.g.

$$\begin{aligned} \beta(v_1 + v_2, w) &= A(v_1 + v_2) \otimes Bw = (Av_1 + Av_2) \otimes Bw \\ &\stackrel{\otimes \text{ bilin.}}{=} Av_1 \otimes Bw + Av_2 \otimes Bw = \beta(v_1, w) + \beta(v_2, w), \end{aligned} \quad (13)$$

and analogously in the second argument. By the universal property there exists a unique linear $\tilde{\beta}$ with $\tilde{\beta}(v \otimes w) := \beta(v, w)$, so that

$$(A \otimes B)(v \otimes w) = Av \otimes Bw. \quad (14)$$

2.3 The Kronecker product as a matrix representation

Let $V = K^m$ and $W = K^n$ and identify (this is *not* a unique representation!) $K^m \times K^n \rightarrow K^{(m \cdot n)}$ with the Kronecker product, i.e. for

$$v = \begin{pmatrix} a_1 \\ \vdots \\ a_m \end{pmatrix} \quad \& \quad w = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} \quad \text{set} \quad v \otimes w := \begin{pmatrix} a_1 w \\ \vdots \\ a_m w \end{pmatrix} = \begin{pmatrix} a_1 b_1 \\ \vdots \\ a_1 b_n \\ \vdots \\ a_m b_1 \\ \vdots \\ a_m b_n \end{pmatrix}. \quad (15)$$

Now choose $\{v_i\}$ to be the standard basis, so that $Av_i = A_{\cdot i}$ (the i -th column of A) and likewise for B . Then

$$\begin{aligned} (A \otimes B)(v_i \otimes v_j) &= (Av_i) \otimes (Bv_j) \stackrel{\text{std. basis}}{=} A_{\cdot i} \otimes B_{\cdot j} \\ &= \begin{pmatrix} A_{1i} B_{\cdot j} \\ \vdots \\ A_{mi} B_{\cdot j} \end{pmatrix} = \begin{pmatrix} \begin{pmatrix} A_{1i} B_{1j} \\ \vdots \\ A_{1i} B_{nj} \end{pmatrix} \\ \vdots \\ \begin{pmatrix} A_{mi} B_{1j} \\ \vdots \\ A_{mi} B_{nj} \end{pmatrix} \end{pmatrix}. \end{aligned} \quad (16)$$

Setting $v_i \otimes v_j \equiv w_k$ with $k = j + n i$ (for zero-based i), the matrix of $A \otimes B$ is the block matrix

$$\begin{aligned} A \otimes B &= \begin{pmatrix} A_{11} B_{\cdot 1} & \cdots & A_{11} B_{\cdot n} & \cdots & A_{1m} B_{\cdot 1} & \cdots & A_{1m} B_{\cdot n} \\ \vdots & & & & & & \vdots \\ A_{m1} B_{\cdot 1} & \cdots & A_{m1} B_{\cdot n} & \cdots & A_{mm} B_{\cdot 1} & \cdots & A_{mm} B_{\cdot n} \end{pmatrix} \\ &= \begin{pmatrix} A_{11} B & \cdots & A_{1m} B \\ \vdots & \ddots & \vdots \\ A_{m1} B & \cdots & A_{mm} B \end{pmatrix}, \end{aligned} \quad (17)$$

which is the Kronecker product for matrices.

2.4 References

- *Tensorprodukt* and *Kronecker-Produkt*, German Wikipedia.

3 The Jordan–Wigner mapping for spinless Fermions

3.1 Summary

The Jordan–Wigner transformation (JWT) can be used to map a fermionic system to a system of spins. This is essentially done by rewriting

$$\text{fermion} = \text{string} \times \text{operator},$$

where the “string” is some product of operators $\hat{Z}_i = 1 - 2\hat{n}_i$ and the “operator” is found to obey spin commutation relations.

As references I used

- *Fermions and Jordan–Wigner String* – ITensor C++ Documentation,
- *Introduction to Many-Body Physics*, Chapter 4.1 – Piers Coleman,
- *The Fermionic canonical commutation relations and the Jordan–Wigner transform* – M. Nielsen,

which I supplemented with some calculations and my own way of structuring things.

3.2 Setup: the fermionic Hilbert space

Our fermionic Hilbert space is defined by (Nielsen):

- i) operators $c_1, \dots, c_L$ acting on some Hilbert space;
- ii) these operators satisfying the canonical (anti-)commutation relations for fermions

$$\{c_i^{(+)}, c_j^{(+)}\} = 0 \quad \& \quad \{c_i, c_j^\dagger\} = \delta_{ij} \mathbb{1}. \quad (18)$$

Many things directly follow from this, such as a possible basis for the Hilbert space, namely the occupation number basis

$$|\vec{n}\rangle = |n_1, n_2, \dots, n_L\rangle \equiv \left(\prod_{i=1}^L (c_i^\dagger)^{n_i} \right) |0\rangle, \quad (19)$$

where $\hat{n}_i |\vec{n}\rangle = c_i^\dagger c_i |\vec{n}\rangle = n_i |\vec{n}\rangle$ with $n_i = 0, 1$ and

$$[\hat{n}_i, \hat{n}_j] = 0. \quad (20)$$

3.3 The transformation

The JWT introduces new operators a_j by writing

$$c_j \equiv \left(\prod_{i=1}^{j-1} Z_i \right) a_j \quad \& \quad c_j^\dagger \equiv a_j^\dagger \left(\prod_{i=1}^{j-1} Z_i \right), \quad (21)$$

with $Z_j = (-1)^{n_j} = 1 - 2n_j$ and $n_j = c_j^\dagger c_j$. The ordering of the Z_i does not matter since $[Z_i, Z_j] = 0$ due to eq. (20). Inverting eq. (21) using $Z_i^2 = \mathbb{1}$ and $[Z_i, Z_j] = 0$ yields

$$a_j = \left(\prod_{i=1}^{j-1} Z_i \right) c_j \quad \& \quad a_j^\dagger = c_j^\dagger \left(\prod_{i=1}^{j-1} Z_i \right). \quad (22)$$

3.4 Algebra of the new operators

One can show that these new operators obey

$$[a_i^{(+)}, a_j^{(+)}] = 0 \quad \forall i, j, \quad (23)$$

$$[a_i, a_j^\dagger] = 0 \quad \forall i \neq j, \quad (24)$$

$$[a_i, a_i^\dagger] = 1 - 2n_i \quad \text{with } n_i = c_i^\dagger c_i, \quad (25)$$

using that

$$\{Z_i, c_j^{(+)}\} = 0 \quad (26)$$

The corresponding proofs can be found in section 3.8.

3.5 The occupation-number basis in the new operators

The basis states of the fermionic Hilbert space can be written as

$$\begin{aligned} |n_1, n_2, \dots, n_L\rangle &\equiv \left(\prod_{i=1}^L (c_i^\dagger)^{n_i} \right) |0\rangle && \text{w. } n_i = 0, 1 \\ &= \left\{ \prod_{i=1}^L \left[a_i^\dagger \left(\prod_{j=1}^{i-1} Z_j \right) \right]^{n_i} \right\} |0\rangle && (c_1^\dagger = a_1^\dagger) \\ &= (a_1^\dagger)^{n_1} (a_2^\dagger Z_1)^{n_2} (a_3^\dagger Z_2 Z_1)^{n_3} \dots (a_L^\dagger Z_{L-1} \dots Z_1)^{n_L} |0\rangle \\ &= \left(\prod_{i=1}^L (a_i^\dagger)^{n_i} \right) (Z_1)^{n_2} (Z_2 Z_1)^{n_3} \dots (Z_{L-1} \dots Z_1)^{n_L} |0\rangle \\ &= \left(\prod_{i=1}^L (a_i^\dagger)^{n_i} \right) |0\rangle && \text{w. } n_i = 0, 1, \end{aligned} \quad (27)$$

using $[Z_i, Z_j] = 0 \ \forall i, j$, $[Z_i, c_j] = 0$ for $i \neq j$ (hence $[Z_i, a_j^\dagger] = 0$ for $i \neq j$), and $Z_j |0\rangle = |0\rangle$. The ordering is not important anymore because $[a_i^\dagger, a_j^\dagger] = 0$.

3.6 Action on basis states

One cannot define the action of a_j ($a_j^\dagger$) locally, since the JW string $Z_1 \cdots Z_{j-1}$ is non-local, so one has to examine them on the full many-body basis. For $n_\ell = 0$,

$$\begin{aligned} a_\ell |n_1, \dots, n_\ell = 0, \dots, n_L\rangle &= a_\ell \left(\prod_{i=1}^L (a_i^\dagger)^{n_i} \right) |0\rangle \\ &\stackrel{[a_\ell, a_j^\dagger]=0}{=} \left(\prod_{i=1}^L (a_i^\dagger)^{n_i} \right) a_\ell |0\rangle = 0, \end{aligned} \quad (28)$$

since $a_\ell |0\rangle = c_\ell |0\rangle = 0$. For $n_\ell = 1$,

$$\begin{aligned} a_\ell |n_1, \dots, n_\ell = 1, \dots, n_L\rangle &= a_\ell \left(\prod_{\substack{i=1 \\ i \neq \ell}}^L (a_i^\dagger)^{n_i} \right) a_\ell^\dagger |0\rangle \\ &\stackrel{[a_\ell, a_j^\dagger]=0}{=} \left(\prod_{\substack{i=1 \\ i \neq \ell}}^L (a_i^\dagger)^{n_i} \right) a_\ell a_\ell^\dagger |0\rangle = \left(\prod_{\substack{i=1 \\ i \neq \ell}}^L (a_i^\dagger)^{n_i} \right) c_\ell c_\ell^\dagger |0\rangle \\ &= \left(\prod_{\substack{i=1 \\ i \neq \ell}}^L (a_i^\dagger)^{n_i} \right) |0\rangle = |n_1, \dots, n_\ell = 0, \dots, n_L\rangle. \end{aligned} \quad (29)$$

Using $[a_\ell^\dagger, a_j^\dagger] = 0$ and $(a_\ell^\dagger)^2 = 0$, one quickly finds

$$a_\ell^\dagger |n_1, \dots, n_\ell = 0, \dots, n_L\rangle = |n_1, \dots, n_\ell = 1, \dots, n_L\rangle, \quad (30)$$

$$a_\ell^\dagger |n_1, \dots, n_\ell = 1, \dots, n_L\rangle = 0. \quad (31)$$

Thus a_ℓ ($a_\ell^\dagger$) destroys (creates) a particle in the many-body state *without* giving rise to an extra sign structure (as c_ℓ ($c_\ell^\dagger$) would do).

3.7 Identification with spins

We can thus identify $n_i = 0, 1$ with $n_i = \downarrow, \uparrow$ in the many-body state, and the new operators can be associated with spins by identifying

$$a_j = S_j^-, \quad a_j^\dagger = S_j^+ \quad \& \quad n_j - \frac{1}{2} = S_j^z = \frac{1}{2} \sigma_j^z, \quad (32)$$

since $[S_j^-, S_j^+] = -2S_j^z$ for spins (see the quick refresher on spin, section 6). For instance

$$a_3^\dagger |0\rangle = S_3^+ |\downarrow\downarrow\downarrow\rangle = |\uparrow\downarrow\downarrow\rangle.$$

Matrix elements do not change under the JWT, since it is only a rewriting of operators and a renaming of basis states.

3.8 Proofs about the JWT

Claim: $\{Z_j, c_j^{(+)}\} = 0$.

Proof. On the single site j , with $Z_j |0\rangle_j = |0\rangle_j$, $Z_j |1\rangle_j = -|1\rangle_j$, $c_j |0\rangle_j = 0$ and $c_j |1\rangle_j = |0\rangle_j$,

$$Z_j c_j |0\rangle_j = Z_j \cdot 0 = 0 = c_j |0\rangle_j = c_j Z_j |0\rangle_j, \quad (33)$$

$$Z_j c_j |1\rangle_j = Z_j |0\rangle_j = |0\rangle_j = c_j |1\rangle_j, \quad (34)$$

$$c_j Z_j |1\rangle_j = -c_j |1\rangle_j = -|0\rangle_j. \quad (35)$$

Hence, on both basis states $\{Z_j, c_j\} = Z_j c_j + c_j Z_j = c_j - c_j = 0$, and since $Z_j^\dagger = Z_j$ it follows that $\{Z_j, c_j^\dagger\} = 0$ as well. (see Coleman, Chapter 4.1) $\square$

Claim: $[a_i^{(+)}, a_j^{(+)}] = 0 \quad \forall i, j$.

Proof. Assume w.l.o.g. that $i < j$. Using eq. (22),

$$\begin{aligned} a_j a_i &= (Z_1 \cdots Z_{j-1} c_j) (Z_1 \cdots Z_{i-1} c_i) \\ &\stackrel{\substack{[Z_j, c_\ell]=0, j \neq \ell \\ \{c_i, c_j\}=0}}{=} Z_1^2 \cdots Z_{i-1}^2 Z_i \cdots Z_{j-1} c_j c_i = -Z_i \cdots Z_{j-1} c_i c_j \quad (= 0 \text{ if } i = j), \end{aligned} \quad (36)$$

and likewise

$$\begin{aligned} a_i a_j &= (Z_1 \cdots Z_{i-1} c_i) (Z_1 \cdots Z_{j-1} c_j) \\ &\stackrel{[Z_i, c_\ell]=0, i \neq \ell}{=} Z_{i+1} \cdots Z_{j-1} c_i Z_i c_j \stackrel{\{Z_i, c_i\}=0}{=} -Z_i \cdots Z_{j-1} c_i c_j. \end{aligned} \quad (37)$$

Comparing the two, $a_i a_j = a_j a_i$, i.e. $[a_i, a_j] = 0$. Analogously for $[a_i^\dagger, a_j^\dagger]$ and for $i > j$; the case $i = j$ is straightforward since $c_i^2 = 0$. $\square$

Claim: $[a_i, a_j^\dagger] = 0 \quad \forall i \neq j$.

Proof. We can show it w.l.o.g. for $i < j$. Using eq. (22) and eq. (26),

$$\begin{aligned} a_i^\dagger a_j &= (c_i^\dagger Z_{i-1} \cdots Z_1) (Z_1 \cdots Z_{j-1} c_j) \stackrel{Z_i^2=1, i < j}{=} c_i^\dagger Z_i \cdots Z_{j-1} c_j \\ &= Z_{i+1} \cdots Z_{j-1} c_i^\dagger Z_i c_j \stackrel{\{Z_i, c_i^\dagger\}=0}{=} -Z_i \cdots Z_{j-1} c_i^\dagger c_j = +Z_i \cdots Z_{j-1} c_j c_i^\dagger, \end{aligned} \quad (38)$$

and

$$\begin{aligned} a_j a_i^\dagger &= (Z_1 \cdots Z_{j-1} c_j) (c_i^\dagger Z_{i-1} \cdots Z_1) \\ &= Z_i \cdots Z_{j-1} c_j c_i^\dagger Z_{i-1}^2 \cdots Z_1^2 \quad ([Z_j, c_\ell^{(+)}] = 0, j \neq \ell) \\ &= Z_i \cdots Z_{j-1} c_j c_i^\dagger. \end{aligned} \quad (39)$$

Both reduce to the same expression, so $a_j a_i^\dagger - a_i^\dagger a_j = 0$ for $i < j$, i.e. $[a_j, a_i^\dagger] = 0$; the case $i > j$ follows straightforwardly. $\square$

Claim: $[a_i, a_i^\dagger] = 1 - 2n_i$ **with** $n_i = c_i^\dagger c_i$.

Proof. Using eq. (22) and $Z_k^2 = \mathbb{1}$,

$$a_i^\dagger a_i = c_i^\dagger Z_{i-1} \cdots Z_1 Z_1 \cdots Z_{i-1} c_i = c_i^\dagger c_i = n_i, \quad (40)$$

$$a_i a_i^\dagger = Z_1 \cdots Z_{i-1} c_i c_i^\dagger Z_{i-1} \cdots Z_1 = c_i c_i^\dagger = 1 - c_i^\dagger c_i, \quad (41)$$

so that

$$[\hat{a}_i, \hat{a}_i^\dagger] = a_i a_i^\dagger - a_i^\dagger a_i = (1 - c_i^\dagger c_i) - c_i^\dagger c_i = 1 - 2\hat{n}_i. \quad (42)$$

□

4 Minimizing complex functions with Wirtinger derivatives

4.1 Definition

Define for $z = x + iy \in \mathbb{C}$ with $x, y \in \mathbb{R}$

$$\frac{\partial}{\partial z} \equiv \frac{1}{2} \left(\frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right) \quad \& \quad \frac{\partial}{\partial \bar{z}} = \frac{1}{2} \left(\frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right) \quad (43)$$

for functions of one complex variable, and

$$\frac{\partial}{\partial z_j} = \frac{1}{2} \left(\frac{\partial}{\partial x_j} - i \frac{\partial}{\partial y_j} \right) \quad \& \quad \frac{\partial}{\partial \bar{z}_j} = \frac{1}{2} \left(\frac{\partial}{\partial x_j} + i \frac{\partial}{\partial y_j} \right) \quad (44)$$

for functions of multiple complex variables.

4.2 Agreement with the complex derivative

If f is complex differentiable at a point (i.e. the limit $f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$ exists), then $\frac{\partial f}{\partial z}$ agrees with the complex derivative $\frac{df}{dz}$.

Proof. For $f(z) = u(z) + iv(z)$ complex differentiable,

$$\begin{aligned} \frac{\partial f}{\partial z} &\stackrel{(43)}{=} \frac{1}{2} \left(\frac{\partial f}{\partial x} - i \frac{\partial f}{\partial y} \right) = \frac{1}{2} \left(\frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} - i \frac{\partial u}{\partial y} + \frac{\partial v}{\partial y} \right) \\ &\stackrel{(43)}{=} \frac{\partial u}{\partial z} + i \frac{\partial v}{\partial z} \stackrel{\text{Def.}}{=} \frac{df}{dz}. \end{aligned} \quad (45)$$

□

4.3 The $\bar{z}$ -derivative and Cauchy–Riemann

Furthermore,

$$\begin{aligned} 0 &\stackrel{!}{=} \frac{\partial f}{\partial \bar{z}} = \frac{1}{2} \left(\frac{\partial f}{\partial x} + i \frac{\partial f}{\partial y} \right) = \frac{1}{2} \left(\frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} + i \frac{\partial u}{\partial y} - \frac{\partial v}{\partial y} \right) \\ &\iff \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \wedge \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}, \end{aligned} \quad (46)$$

so that $0 = \frac{\partial f}{\partial \bar{z}}$ is equivalent to the Cauchy–Riemann equations, i.e. the necessary & sufficient condition for f to be complex differentiable.

4.4 Minimizing a real-valued function

Minimize a real-valued function¹

$$f: \mathbb{C} \rightarrow \mathbb{R} \tag{47}$$

$$f(z) = f(x, y), \quad (z = x + iy) \tag{48}$$

via the condition

$$\nabla f(x, y) = \begin{pmatrix} \partial_x f \\ \partial_y f \end{pmatrix} \stackrel{!}{=} 0. \tag{49}$$

Then

$$\partial_x f \stackrel{!}{=} 0 \stackrel{!}{=} \partial_y f \quad \Longleftrightarrow \quad \frac{\partial f}{\partial \bar{z}} = \frac{1}{2} \left(\frac{\partial f}{\partial x} + i \frac{\partial f}{\partial y} \right) \stackrel{!}{=} 0, \tag{50}$$

so one can minimize a real-valued function by imposing $\boxed{\frac{\partial f}{\partial \bar{z}} = 0}$.

4.5 References

- *Wirtinger derivatives*, English Wikipedia.

¹Minimizing a complex-valued f does not make sense because $\mathbb{C}$ has no order.

5 The density matrix, entanglement entropy and their connection to singular value decomposition (SVD)

5.1 Main goal

Prove that for λ_i the eigenvalues of the reduced density matrix and s_i the Schmidt values,

$$S_{\text{Neum}} \equiv \text{Tr}(\rho_A \log \rho_A) = \sum_i \lambda_i \log \lambda_i = \sum_i s_i^2 \log s_i^2. \quad (51)$$

5.2 The density operator

Let $\{p_i, |\psi_i\rangle\}$ be an ensemble of pure states, describing a system that is in one of the states $|\psi_i\rangle$ with probability p_i , i.e. the system is not fully known. Define the density operator as

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|, \quad p_i \in \mathbb{R} \ \& \ \sum_i p_i = 1, \quad (52)$$

which is a positive semi-definite operator of trace one (see for example Nielsen & Chuang).

5.3 Observables on a composite system

Take a system A with some observable M , and a system B , giving the composite system AB (see Box 2.6). Assume we can measure M on system A ; then a corresponding measurement on AB is $\tilde{M} = M \otimes \mathbb{1}_B$.

Proof. Take AB to be prepared in $|m\rangle|\psi\rangle$ with $M|m\rangle = m|m\rangle$. Applying the measurement to $|m\rangle|\psi\rangle$ should yield m with probability one, i.e. the corresponding projector for $\tilde{M}$ is $P_m \otimes \mathbb{1}_B$. By the spectral theorem, $\tilde{M} = \sum_m m P_m \otimes \mathbb{1}_B = M \otimes \mathbb{1}_B$ for a general state of AB . $\square$

5.4 The reduced density matrix as a partial trace

Measurement averages must be the same whether computed via the full ρ^{AB} or via some ρ^A associated to system A , i.e.

$$\text{tr}(M\rho^A) \stackrel{!}{=} \text{tr}(\tilde{M}\rho^{AB}) = \text{tr}((M \otimes \mathbb{1}_B)\rho^{AB}). \quad (53)$$

Take $\rho^A = \text{tr}_B(\rho^{AB})$ where

$$\text{tr}_B(|a_1\rangle\langle a_2| \otimes |b_1\rangle\langle b_2|) \equiv |a_1\rangle\langle a_2| \text{tr}(|b_1\rangle\langle b_2|) \quad \& \quad \text{tr}_B(\cdot) \text{ linear.} \quad (54)$$

Write ρ^{AB} by using the eigenbasis of M for system A with $M|i\rangle = m_i|i\rangle$ and $\langle i|j\rangle = \delta_{ij}$:

$$\rho^{AB} = \sum_\alpha p^\alpha |\psi\rangle\langle\psi|^\alpha = \sum_\alpha p^\alpha \sum_{ijkl} c_{ij}^\alpha (c_{k\ell}^\alpha)^* |i\rangle_A |j\rangle_B \langle\ell|_B \langle k|_A, \quad (55)$$

$$\rho^A = \sum_\alpha \sum_{ijkl} p^\alpha c_{ij}^\alpha (c_{k\ell}^\alpha)^* |i\rangle\langle k|_A \underbrace{\text{tr}(|j\rangle\langle\ell|_B)}_{=\delta_{j\ell}} = \sum_{\alpha, ijk} p^\alpha c_{ij}^\alpha (c_{kj}^\alpha)^* |i\rangle\langle k|_A. \quad (56)$$

Therefore

$$\text{tr}(M\rho^A) = \sum_{\alpha,ijk} p^\alpha c_{ij}^\alpha (c_{kj}^\alpha)^* \underbrace{\text{tr}(m_i |i\rangle\langle k|_A)}_{=m_i \delta_{ik}} = \sum_{\alpha,ij} p^\alpha m_i c_{ij}^\alpha (c_{ij}^\alpha)^*, \quad (57)$$

and similarly $\text{tr}((M \otimes \mathbb{1}_B)\rho^{AB}) = \sum_{\alpha,ij} p^\alpha m_i c_{ij}^\alpha (c_{ij}^\alpha)^*$, which proves the claim. One can also easily show that this is the only map that conserves consistency (see Box 2.6).

5.5 Reduced density matrix of a pure state

Now examine the reduced density matrix for a pure state,

$$\rho_{AB} = |\psi\rangle\langle\psi| = \sum_{ijk\ell} c_{ij}(c_{k\ell})^* |i\rangle_A |j\rangle_B \langle\ell|_B \langle k|_A, \quad (58)$$

$$\rho_A = \text{tr}_B(\rho_{AB}) = \sum_{ijk} c_{ij}(c_{kj})^* |i\rangle\langle k|_A, \quad (59)$$

such that the matrix representation is

$$\boxed{\rho_A = \psi\psi^\dagger}, \quad (60)$$

using that the c_{ij} parametrize the matrix $\psi \in \mathbb{C}^{d_A \times d_B}$, i.e. $|\psi\rangle = \sum_{ij} c_{ij} |i\rangle_A |j\rangle_B$.

5.6 Connection to the SVD

Now use the SVD on ψ :

$$\psi = USV^\dagger \quad \text{with } U \in \mathbb{C}^{d_A \times d_A}, V \in \mathbb{C}^{d_B \times d_B} \text{ unitary.} \quad (61)$$

Then, using $V^\dagger V = \mathbb{1}$,

$$\rho_A = USV^\dagger V S U^\dagger = US^2 U^\dagger, \quad S^2 = \text{diag}(s_1^2, s_2^2, \dots) \in \mathbb{R}_{\geq 0}^{d_A \times d_A}. \quad (62)$$

Furthermore, since ρ_A is hermitian we can diagonalize it with $\rho_A = O D O^\dagger = US^2 U^\dagger$ and $D = \text{diag}(\lambda_1, \dots)$, so that

$$\boxed{\lambda_i = s_i^2}, \quad (63)$$

i.e. the eigenvalues are given by the (squared) Schmidt values.

5.7 Functions of the density matrix via the spectral theorem

Since ρ_A is hermitian, we can use the spectral theorem to write it as $\rho_A = \sum_i \lambda_i P_i$ where λ_i are its eigenvalues and the eigenvectors $|u_i\rangle$ give $P_i = |u_i\rangle\langle u_i|$, since

$$\rho_A = U D U^\dagger = (|u_1\rangle, \dots, |u_n\rangle) \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} \begin{pmatrix} \langle u_1| \\ \vdots \\ \langle u_n| \end{pmatrix}, \quad \langle u_i | u_j \rangle = \delta_{ij}. \quad (64)$$

Thus a function $f(\cdot)$ applied to ρ_A gives

$$\begin{aligned} f(\rho_A) &= \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} \rho_A^n \stackrel{(*)}{=} \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} \sum_i \lambda_i^n |u_i\rangle\langle u_i| \\ &= \sum_i \left(\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} \lambda_i^n \right) |u_i\rangle\langle u_i| = \sum_i f(\lambda_i) |u_i\rangle\langle u_i| = U \text{diag}(f(\lambda_i)) U^\dagger, \end{aligned} \quad (65)$$

where $(*)$ uses $\rho_A^n = \sum_i \lambda_i^n |u_i\rangle\langle u_i|$ (e.g. $\rho_A \rho_A = \sum_i \lambda_i^2 |u_i\rangle\langle u_i|$).

5.8 The Von Neumann entanglement entropy

Therefore, for the Von Neumann entanglement entropy,

$$\begin{aligned} S &= \text{Tr}(\rho_A \log \rho_A) = \text{Tr}(\rho_A U \log(\text{diag}(\lambda_i)) U^\dagger) \\ &= \text{Tr}(U \text{diag}(\lambda_i) \underbrace{U^\dagger U}_{=1} \log(\text{diag}(\lambda_i)) U^\dagger) \\ &= \sum_i \lambda_i \log \lambda_i = \boxed{\sum_i s_i^2 \log s_i^2}. \end{aligned} \quad (66)$$

S is zero for two unentangled systems and non-zero for entangled systems.

5.9 Example

In our case, for example, $L = 2$, $N_f = 1$, $N_{\text{ph}} = 1$:

$$\{|01\rangle, |10\rangle\} \ \& \ \{|0\rangle, |1\rangle\} \quad \text{as bases of } \mathcal{H}_f \ \& \ \mathcal{H}_{\text{ph}}, \quad (67)$$

$$\Rightarrow \{|01, 0\rangle; |01, 1\rangle; |10, 0\rangle; |10, 1\rangle\} \quad \text{as basis of } \mathcal{H}_f \otimes \mathcal{H}_{\text{ph}}, \quad (68)$$

$$\Rightarrow |\psi\rangle = \sum_{i=1}^2 \sum_{j=1}^2 \psi_{ij} |i\rangle_f |j\rangle \quad \text{with } (\psi)_{ij} = \psi_{ij} \ \& \ \psi \in \mathbb{C}^{2 \times 2}, \quad (69)$$

$$\Rightarrow S(\rho_A) = \sum_{\ell} s_{\ell}^2 \log s_{\ell}^2. \quad (70)$$

Note that since $\text{tr } \rho^{AB} = \text{tr } \rho^A = \text{tr } \rho^B = 1$ we must have $\sum_{\ell} \lambda_{\ell} = \sum_{\ell} s_{\ell}^2 = 1$.

5.10 References

- M. Nielsen & I. Chuang, *Quantum Computation and Quantum Information* (density operator, partial trace; Box 2.6).
- U. Schollwöck, *The density-matrix renormalization group in the age of matrix product states*, beginning of Chapter 4 (MPS).

6 Quick refresher on spin

Spin can be described with the operators $\hat{S}_x, \hat{S}_y, \hat{S}_z$ that obey the following commutation relations:

$$\boxed{[\hat{S}_j, \hat{S}_k] = i\hbar \epsilon_{jkl} \hat{S}_\ell} \quad (\hbar = 1), \quad (71)$$

with $i, j, k = x, y, z = 1, 2, 3$ and using the Einstein summation convention. One can define raising and lowering operators as

$$\boxed{\hat{S}_\pm = \hat{S}_x \pm i\hat{S}_y} \quad \text{with} \quad \boxed{[S_-, S_+] = -2S_z}. \quad (72)$$

Proof.

$$\begin{aligned} [S_-, S_+] &= [S_x - iS_y, S_x + iS_y] \\ &= [S_x, S_x] + i[S_x, S_y] - i[S_y, S_x] + [S_y, S_y] \\ &= 2i[S_x, S_y] \stackrel{(71)}{=} 2i \cdot i \epsilon_{12\ell} S_\ell = -2 \epsilon_{12\ell} S_\ell = -2S_z, \end{aligned} \quad (73)$$

where the first step used the bilinearity of the commutator and the last step uses $\epsilon_{jkl} = 0$ if two indices are equal, leaving $\epsilon_{123} S_3 = S_z$. $\square$

For spin- $\frac{1}{2}$ particles, one can write

$$S_j = \frac{\hbar}{2} \sigma_j \quad \text{with } \sigma_j \text{ the Pauli matrices, } j = x, y, z, \quad (74)$$

and often again $\hbar = 1$.

7 Quick refresher on unitary transformations

Let's say we have some Hamiltonian $\hat{H}$, which we examine in a basis $\{|v_j\rangle\}$ with $j = 1, \dots, \dim(\mathcal{H})$ and $\hat{H}: \mathcal{H} \rightarrow \mathcal{H}$. The matrix representation of the operator $\hat{H}$ in this basis is thus given by

$$H_{ij} = \left\langle v_i \left| \hat{H} v_j \right. \right\rangle = \langle v_i | \hat{H} | v_j \rangle. \quad (75)$$

Applying a unitary transformation U (with $U^\dagger U = \mathbb{1} = UU^\dagger$), i.e.

$$|v_j\rangle \rightarrow |\tilde{v}_j\rangle = U |v_j\rangle \quad \& \quad \hat{H} \rightarrow \hat{\tilde{H}} = U \hat{H} U^\dagger, \quad (76)$$

yields the matrix elements

$$\tilde{H}_{ij} = \langle \tilde{v}_i | \hat{\tilde{H}} | \tilde{v}_j \rangle = \langle v_i | \underbrace{U^\dagger U}_{=\mathbb{1}} \hat{H} \underbrace{U^\dagger U}_{=\mathbb{1}} | v_j \rangle = \langle v_i | \hat{H} | v_j \rangle = H_{ij}. \quad (77)$$

Hence the matrix representation of an operator does *not* change under a unitary transformation.

8 Quick refresher on the current in electrodynamics

8.1 Maxwell's equations

Given Maxwell's equations,

$$\vec{\nabla} \cdot \vec{E}(\vec{r}, t) = \frac{1}{\varepsilon_0} \rho(\vec{r}, t) \quad (78)$$

$$\vec{\nabla} \cdot \vec{B}(\vec{r}, t) = 0 \quad (79)$$

$$\vec{\nabla} \times \vec{E}(\vec{r}, t) = -\frac{\partial}{\partial t} \vec{B}(\vec{r}, t) \quad (80)$$

$$\vec{\nabla} \times \vec{B}(\vec{r}, t) = \frac{1}{c^2} \frac{\partial}{\partial t} \vec{E}(\vec{r}, t) + \frac{1}{\varepsilon_0 c^2} \vec{j}(\vec{r}, t) \quad (81)$$

8.2 Continuity equation

From $\varepsilon_0 \partial_t$ (eq. (78)), using $\partial_x \partial_t = \partial_t \partial_x$,

$$\frac{\partial}{\partial t} \rho(\vec{r}, t) = \varepsilon_0 \vec{\nabla} \cdot (\partial_t \vec{E}(\vec{r}, t)), \quad (82)$$

and from $\varepsilon_0 c^2 \vec{\nabla} \cdot$ (eq. (81)),

$$\begin{aligned} \vec{\nabla} \cdot \vec{j}(\vec{r}, t) &= \vec{\nabla} \cdot (\vec{\nabla} \times \vec{B}(\vec{r}, t)) - \varepsilon_0 \vec{\nabla} \cdot (\partial_t \vec{E}(\vec{r}, t)) \\ &= -\varepsilon_0 \vec{\nabla} \cdot (\partial_t \vec{E}(\vec{r}, t)) \quad (\text{div}(\text{rot } \vec{F}) = 0). \end{aligned} \quad (83)$$

Adding the two gives the continuity equation

$$\frac{\partial}{\partial t} \rho(\vec{r}, t) + \vec{\nabla} \cdot \vec{j}(\vec{r}, t) = 0. \quad (84)$$

8.3 Point-particle solutions

Solutions of the continuity equation:

$$\rho(\vec{r}, t) = \sum_{\alpha} q_{\alpha} \delta(\vec{r} - \vec{r}_{\alpha}(t)) \quad (85)$$

$$\vec{j}(\vec{r}, t) = \sum_{\alpha} q_{\alpha} \vec{v}_{\alpha}(t) \delta(\vec{r} - \vec{r}_{\alpha}(t)) \quad (86)$$

Proof. Note that

$$\begin{aligned} \partial_t \delta(\vec{r} - \vec{r}_{\alpha}(t)) &= \partial_t \left(\delta(r_x - r_x^{\alpha}(t)) \delta(r_y - r_y^{\alpha}(t)) \delta(r_z - r_z^{\alpha}(t)) \right) \\ &= - \sum_{i=x,y,z} \dot{r}_i^{\alpha}(t) \partial_{r_i} \delta(r_i - r_i^{\alpha}(t)) = -\dot{\vec{r}}_{\alpha}(t) \cdot \vec{\nabla} \delta(\vec{r} - \vec{r}_{\alpha}(t)), \end{aligned} \quad (87)$$

where we used the chain rule in the second line and

$$\begin{aligned}
\vec{\nabla} \cdot \left(\vec{v}_\alpha(t) \delta(\vec{r} - \vec{r}_\alpha(t)) \right) &= \sum_{i=x,y,z} \partial_{r_i} \left(v_i^\alpha(t) \delta(\vec{r} - \vec{r}_\alpha(t)) \right) \\
&= \sum_{i=x,y,z} v_i^\alpha(t) \partial_{r_i} \delta(\vec{r} - \vec{r}_\alpha(t)) = \vec{v}_\alpha(t) \cdot \vec{\nabla} \delta(\vec{r} - \vec{r}_\alpha(t)).
\end{aligned} \tag{88}$$

Therefore

$$\frac{\partial}{\partial t} \rho(\vec{r}, t) = - \sum_{\alpha} q_{\alpha} \dot{\vec{r}}_{\alpha}(t) \cdot \left(\vec{\nabla} \delta(\vec{r} - \vec{r}_{\alpha}(t)) \right) = - \vec{\nabla} \cdot \vec{j}(\vec{r}, t), \tag{89}$$

which is (84). □

Note that these solutions are not unique (one equation for four components).